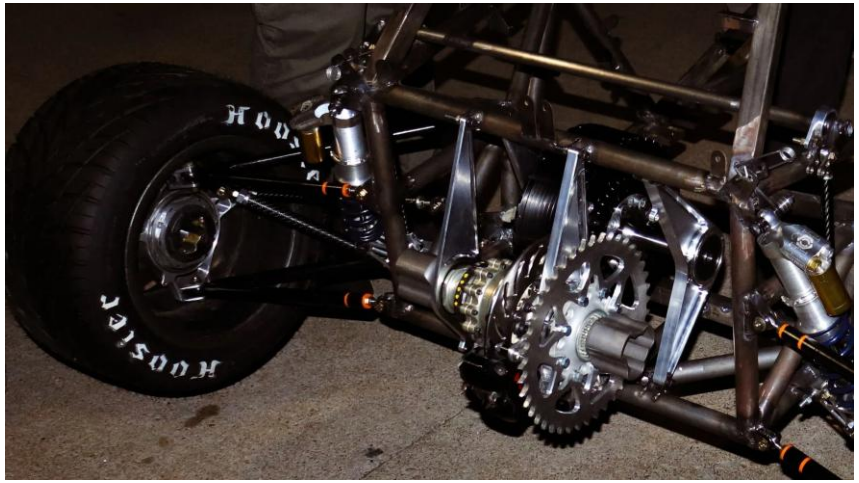
LHRE



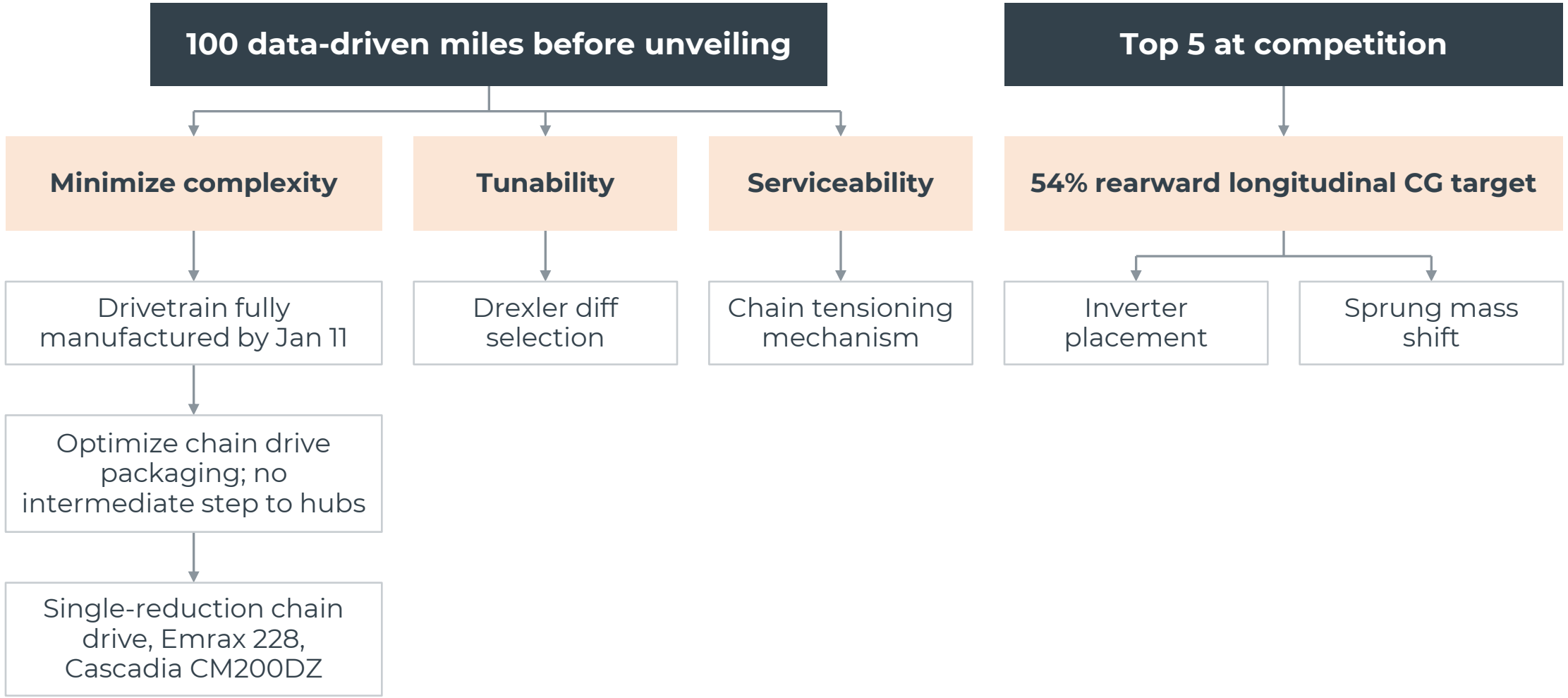


Drivetrain

Ella, Caleb, Elm

ELLA SHEPHERD
Drivetrain Lead

High-Level Goals



LHRE

Architecture & Packaging

ELLA SHEPERD
Drivetrain Lead

Motor & Inverter Selection

- Running back Single Motor Chain RWD
- Emrax 228 + Cascadia CM200DZ minimizes complexity
  - Team has experience with it -> saves integration time
- Downsizing to Emrax 208 would limit max speed

Differential Selection

Team goal is 100 data driven miles

-> How does this apply to drivetrain?

- Data driven for drivetrain means we want to prioritize tunability
- Therefore, we want to run an LSD
- Independent tunability of accel/decel in Drexler and braking through diff were deciding factors over Mk3

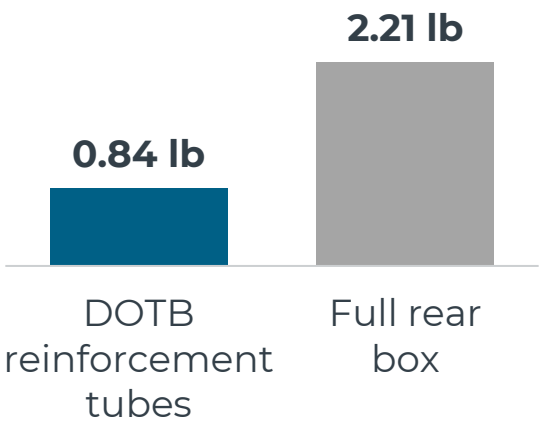
Components (lbs)	Drexler V2	Quiafe MK3	Drexler V2+ custom parts
Differential	5.25	n/a	5.25
Sprocket adapter (Orion)	0.3	n/a	0
bearings	0.24	n/a	0.24
Stub shafts	2.7	n/a	1 (Aluminum est.)
total	8.49	7.1	6.5

Diff off the Back Trade Study

Running diff off the back saves a minimum of 1.38 lbs in frame weight, which outweighs potential drivetrain/suspension benefits

Rules request: a connected motor/diff mount counts as one non-crushable object; 100 mm battery clearance would be measured from the motor

Added rear-box mass by architecture



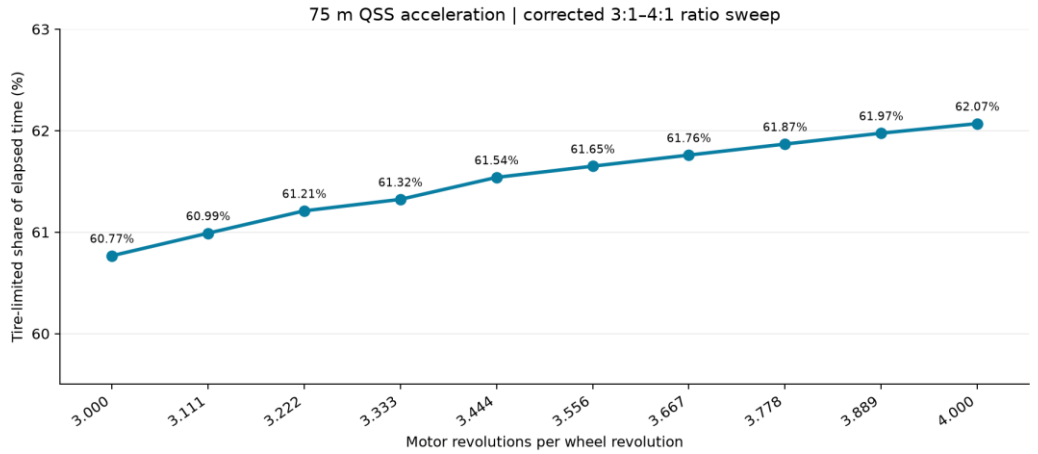
+1.38 lb
(163%) rear box mass increase without DOTB

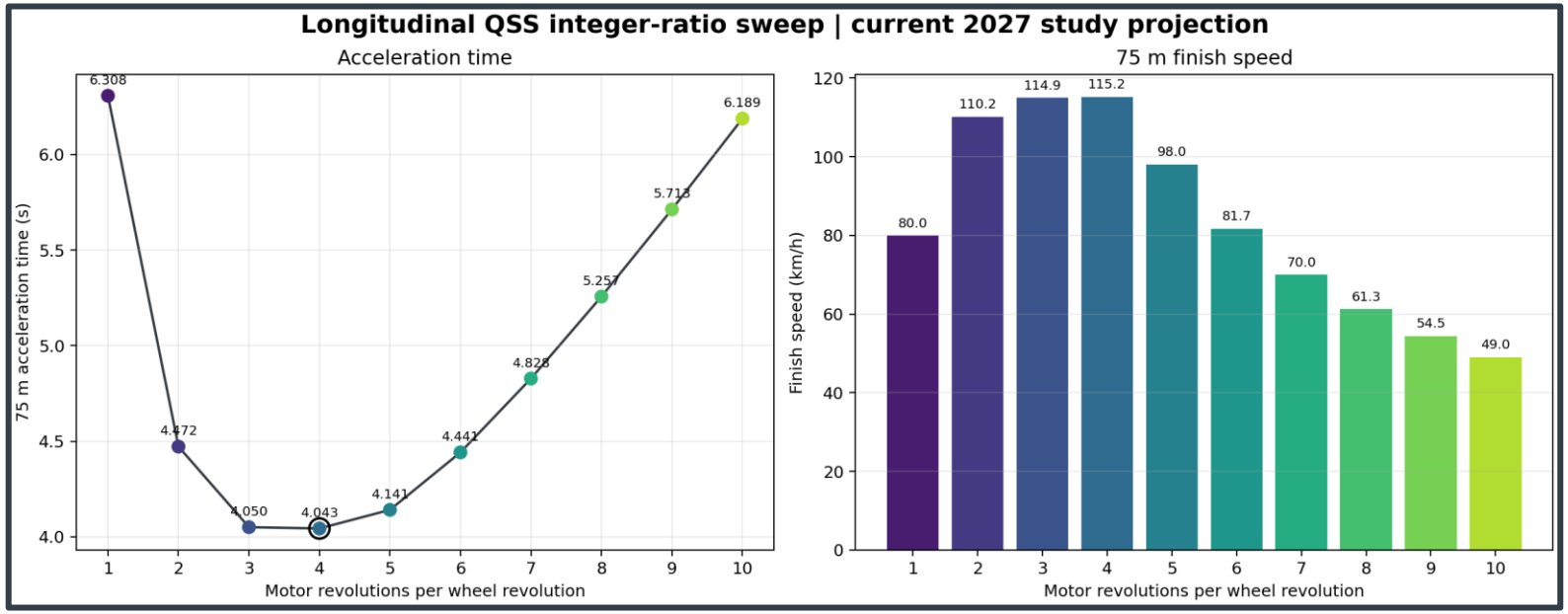
Assumes no improvement in DOTB drivetrain reinforcement tubes, and no reinforcement tubes needed for non-DOTB

Option	Status	Deciding factor
Diff off the back (DOTB)		
A Axial member, size B tube (<100 mm)	Rejected	Sizing up to B tubes is equivalent to adding 8 size C reinforcement tubes
B Axial member, size C tube (>100 mm)	Rejected	Half shaft angle gets cooked (nominal 19.68 degrees)
C No axial member, Angelique type (four mounts)	**Selected**	Avoids A's tube/mount weight and B's halfshaft angle
Non Diff off the back		
D Axially closed drive unit	Rejected	Adds 1.38 lbs rear box mass

Gear Ratio Sizing

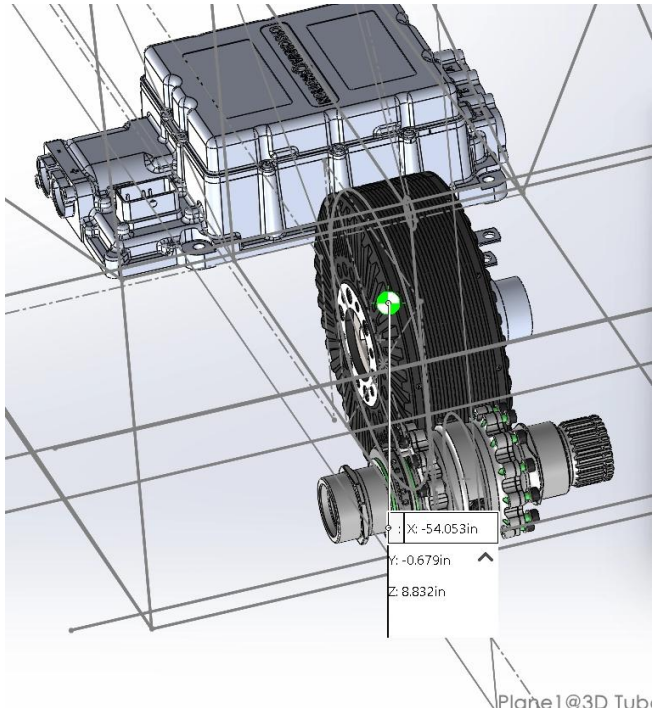
- 43:13 **(3.3)**
- Lower gear ratio -> smaller driven sprocket
  - Less loads on your bearings and sprocket
  - Less mass
  - Smaller Packaging



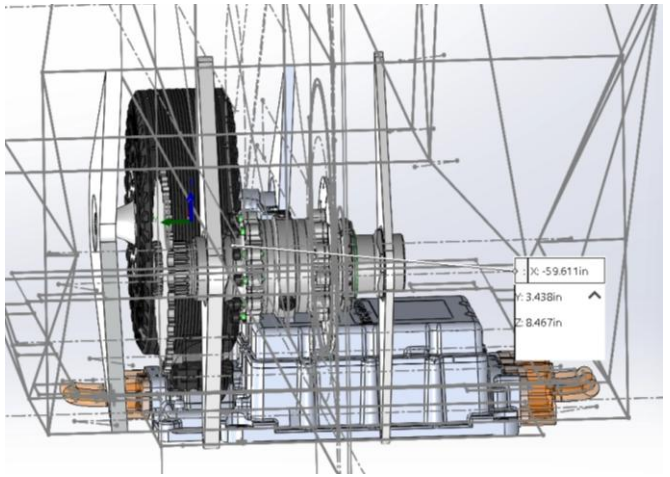


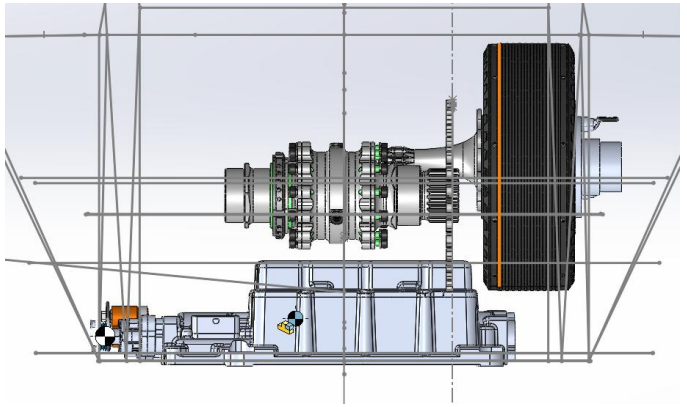
Space Claim Packaging

Orion Style



Floor inverter





Architecture Concepts

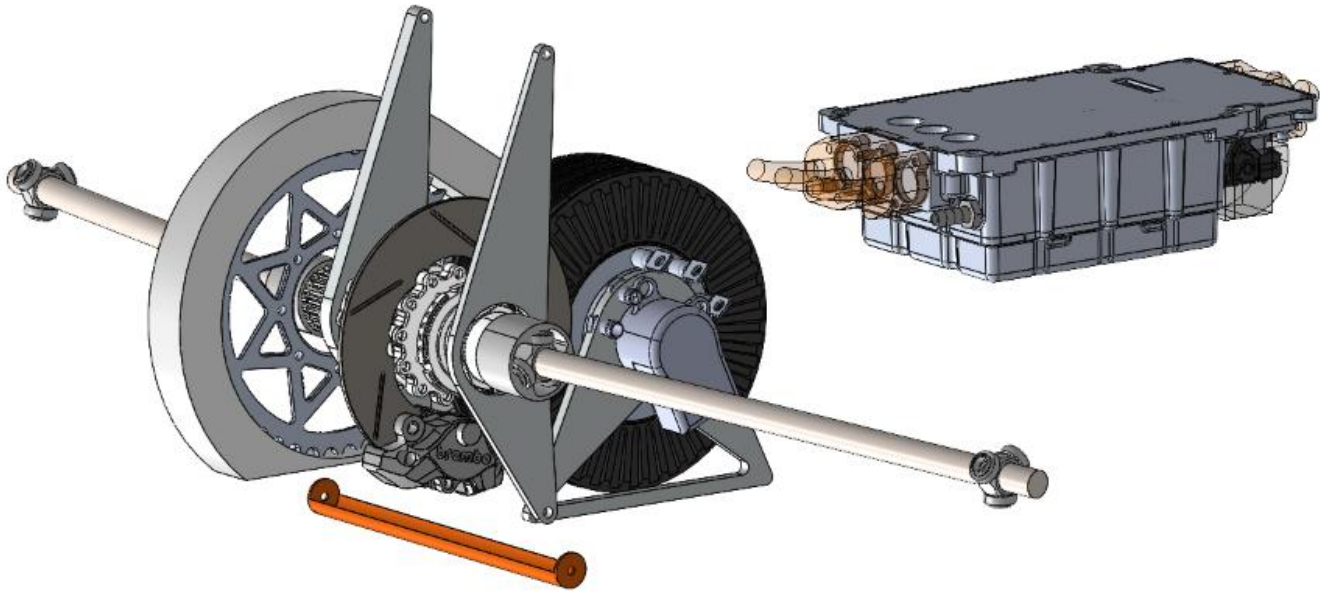
- Floor vs. Orion inverter mounting, equally weighted

config	Car wide CG delta	Combined CG points
Floor inverter	-4.9 mm	1.737
Orion Style	-2.88 mm	1.017
Orion Style-- Flipped Inverter	-3.95 mm	1.40

Criterion	Floor	Pts	Orion	Pts
Mounts	no inverter mounts	1	carbon (complexity)	0
Sprocket		0	center driven sprocket possible	1
Halfshafts	One angle will be higher -> not as far diff position for same max angle; torque steer	0	further diff pos, with max angle	1
Suspension packaging	wider rear box	0	Can shrink wrap	1
Jacking point	(off center diff -> could use frame tube)	1	Have to make a separate one	0
CG		1	Higher and more forward	0
Chain guard	have to wrap it all the way around chain	0		1
Frame	no reinforcement tubes but have to extend	0.5	can shrink wrap	0.5
Risk		0.5	familiar	1
Total		**4**		**5.5**

Final Architecture Selection

- Model correlated CG is worth 9 points per inch of CG z height
  - 3.95 mm z height decrease
- Sprung mass shift achieved 54% rearward bias
- Inverter flip helps with mounting and CG



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Assembly/Part Design

:PPPPPPP

Drivetrain gang

Load Cases, Cycles, and Safety Factors

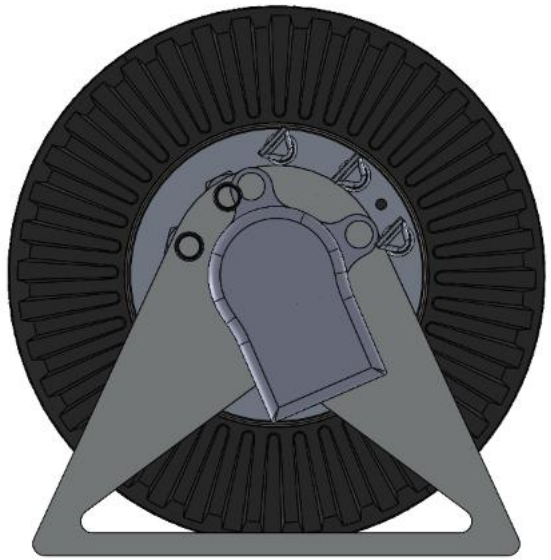
- 2x Safety Factor
- Spec for 30 endurances
- 220 Nm peak torque * 43/13 gear ratio
- 1.5 chain shock factor
- 27,000 shock cycles (1 per endurance)

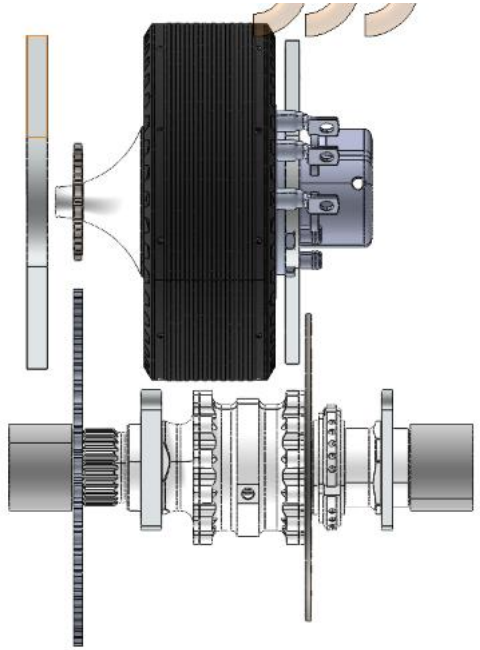
Wheel radius	8 in
Wheel circumference	50.27 in
Distance (30 endurances)	410 mi
Total distance	25,977,600 in
Diff assembly cycles	516,808
Motor assembly cycles	1,709,442

Motor max speed	6,500 rpm
Diff max speed	1,965 rpm

Motor Assembly

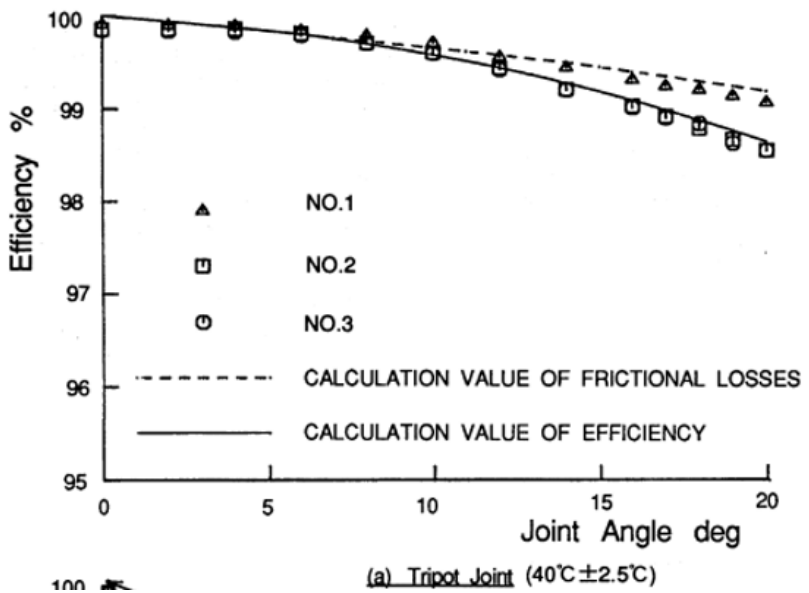
- Triangle type motor mount design to stiffen frame and delete steel reinforcement tubes
- Mounting brake rotor on preload nut side avoids a cantilevered MOS
  - Making custom pins
  - Mass: 359g (2026) vs. 292g (2024) -> ~19% decrease
  - Making pins not deep, better to save mass through architecture

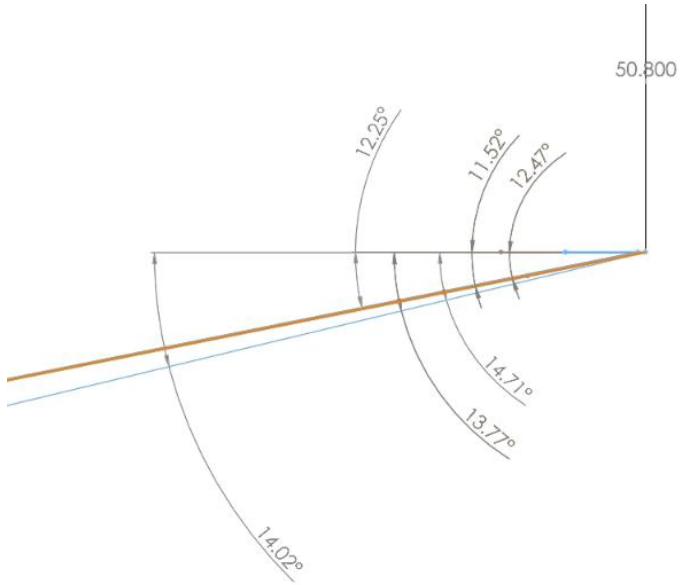




Differential Assembly

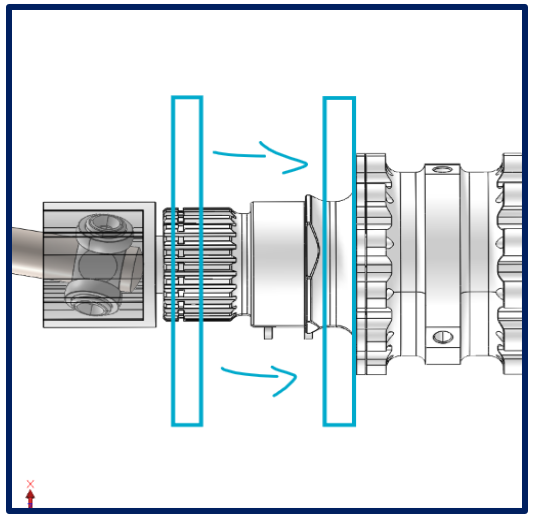
- TRE Lightning series: “Max angle of ~25 deg. Deeper plunge will reduce max angle”
- Weight savings in tripod bearing (0.37 lbs to 0.25 lbs -> 0.48 lbs) and halfshafts (OD 20 -> 18mm)
  - CG bias > efficiency losses





Center Driven Sprocket?

- Pros
  - Improved loading on tabs
  - Scrap spline-adapter
- Cons
  - Uncertainties
  - Massive failure case
  - More complex

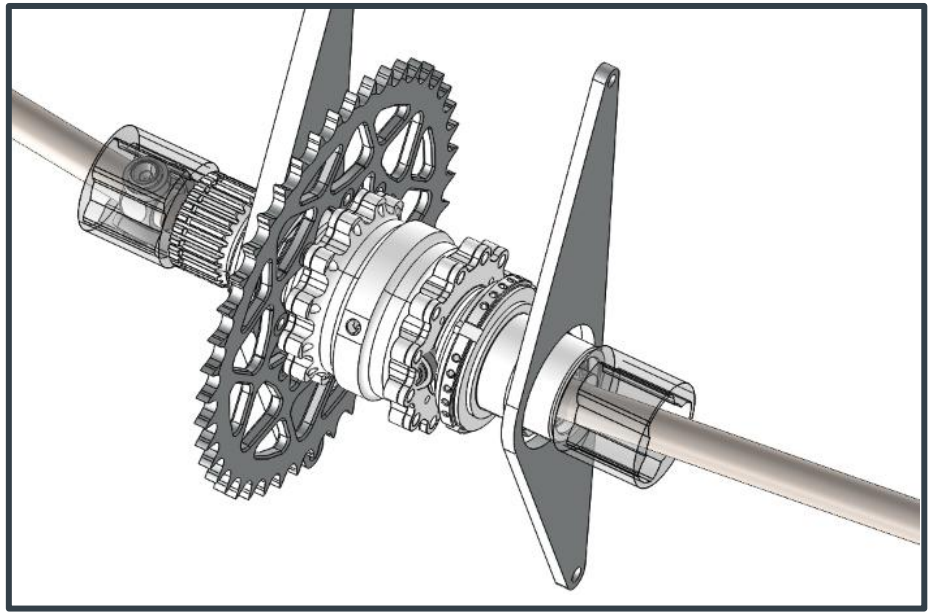


Side driven

	Upper Tab	Lower Tabs
Left Total Load	4558.841562	7635.068098
Right Total Load	776.0190236	1299.663085

Center driven

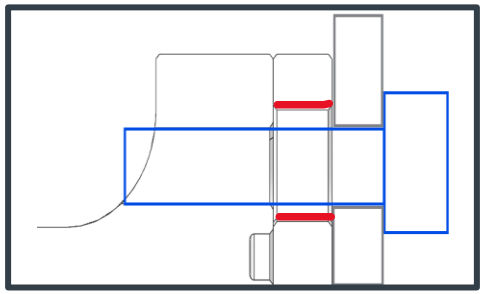
	Upper Tab	Lower Tabs
Left Total Load	3238.366295	5423.55922
Right Total Load	544.4562439	911.8457929

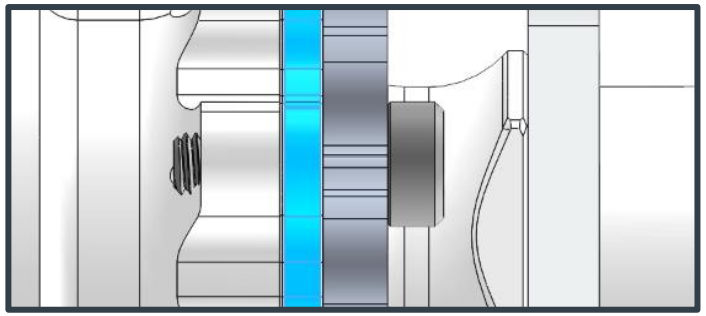


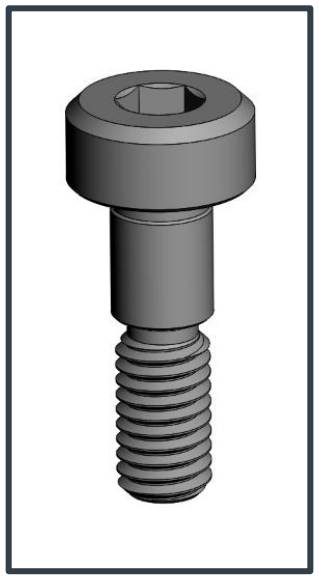
	Loading	Simplicity	Certainty	Failure magnitude	Weighted score
center driven	0.7	0.3	0	.2	0.23
non-center driven	0.3	0.7	1	.8	0.77
weight	10	20	20	50	

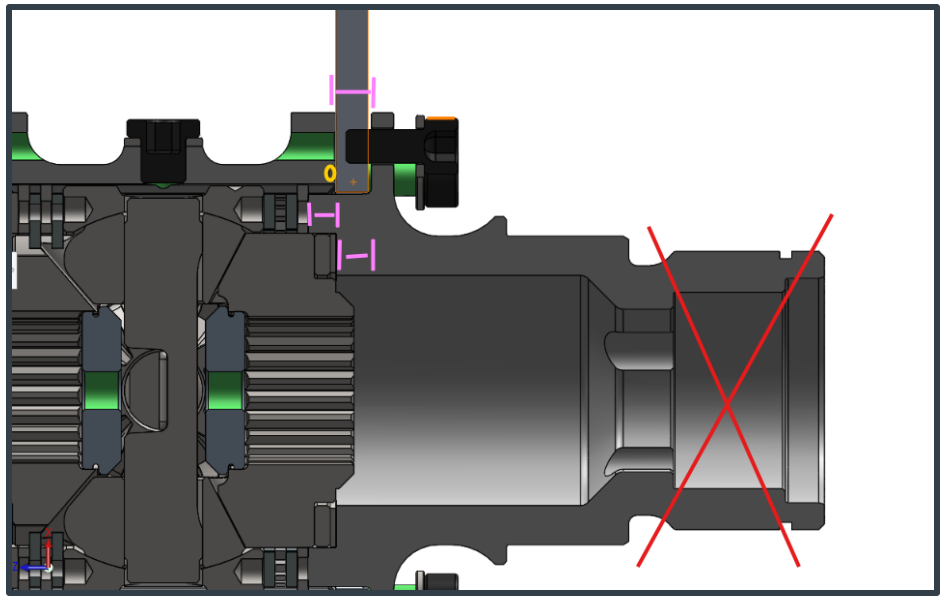
Center driven cont.

- Bolt is in bending b/c of clearance hole that cant react the sprocket
- Solution: Remove bending
  - Custom Bolts – add perfect fit shoulder
  - Custom Diff Part – remove the clearance hole
  - Sprocket in between diff?





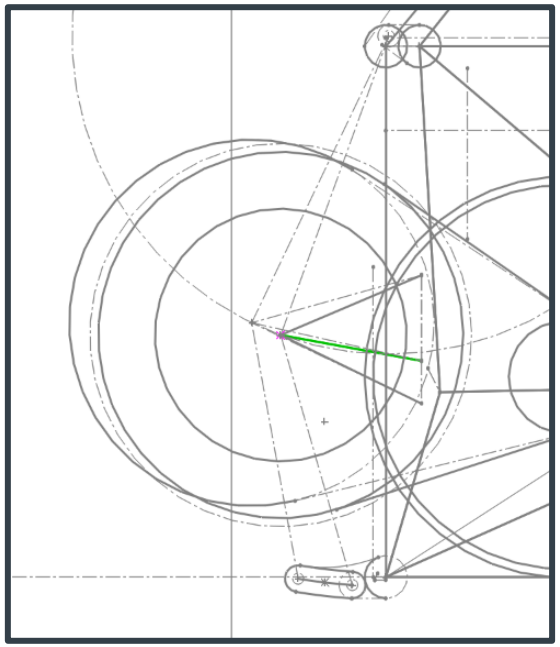


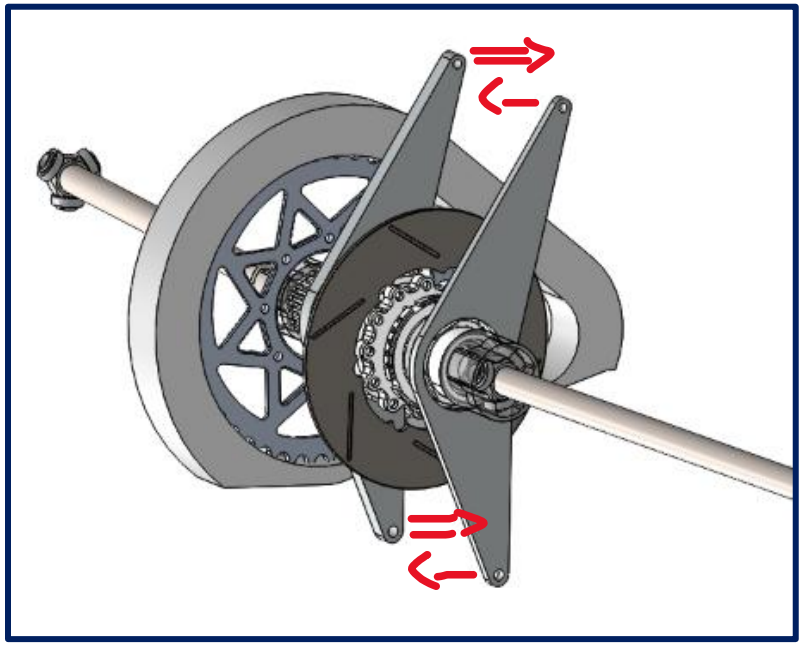


Torque(Nm)	Shear Force (N)	Shear Force (N)
10	2925	2925
Nut Factor	**Length (mm)**	**Area (mm^2)**
0.26	7.175	20.1
Preload (N)	**y (mm)**	
6410.25641	2.3865	
Area (mm^2)	**I**	
20.1	25.47625787	
Preload (Mpa)	**Bending (Mpa)**	**Shear (MPa)**
318.9182294	1965.955025	145.5223881
Tensile (Mpa)		**Equivalent stress (Mpa)**
318.9182294		406.4961664
		Equivalent Stress+Frict.
		348.9305131
Remaining Friction	**SF**	**SF w bending**
81.73874219	3.095172132	0.471769155

Diff Mounts

- Rules: Tangent to top & bottom of rear bulkhead
- Thickness determined by bearings (sizing tbd)
- Swept for 2 pitch lengths
  - For infinite adjustability
- Chain Tensioning Mech:
  - Extensions removes slip case
  - Easy to manufacture and allows for consistent & measurable tension



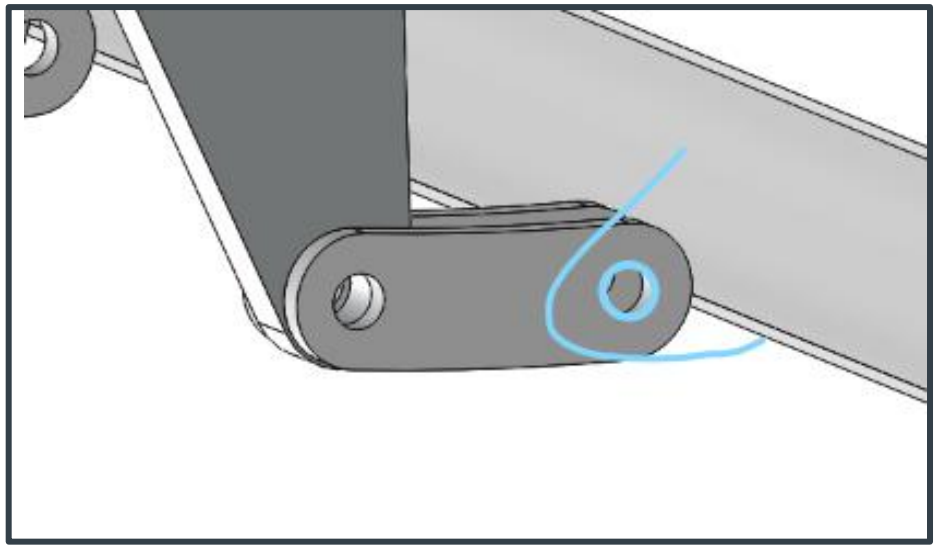


In rear view, the Rear Bulkhead must protect the Tractive System components with a diagonal structure or X brace that meets **F.11.3.1 above**

a. Differential mounts, two vertical tubes with similar spacing, a metal plate, or an Equivalent composite panel may replace a diagonal

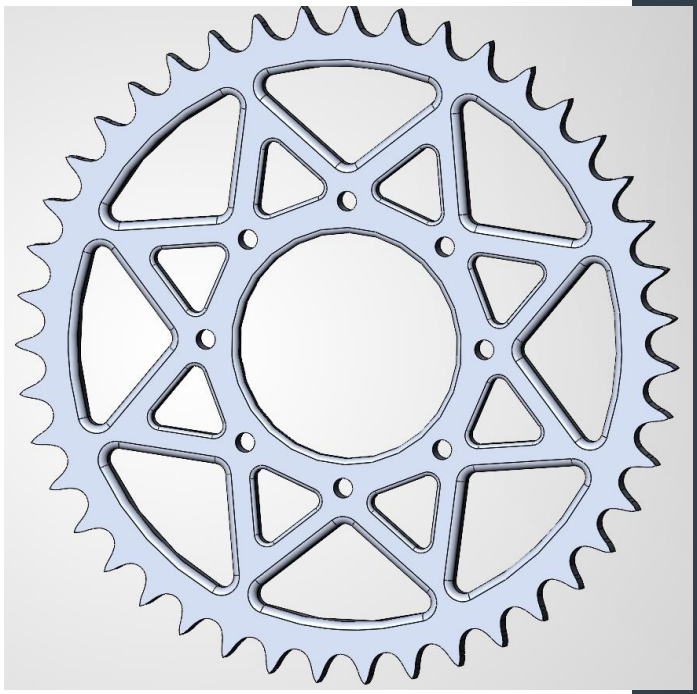
If used, the mounts, plate, or panel must:

- Be aft of the upper and lower Rear Bulkhead structures
- Go to the top and bottom edges of the Rear Bulkhead structures



Sprocket

- Constraints/Requirements:
  - Spec for 30 endurance runs
    - 30 endurance runs = 22km*30 = 66000m
    - 1 revolution = π*Wheel Diameter = π*.4064m
    - Total revs = 66000m/(π*.4064m) = ~51694 revs
  - Motor Peak Torque = 220Nm
- Future Plans:
  - Get forces on sprocket by figuring out this equation for back tension on tooth k $T_k = T_0 * \left(\sin(\varphi) / \left(\sin\varphi + \frac{360}{N} \right) \right)^{k-1}$
  - ^For future FEA testing
- Future Ideas to Research:
  - Asymmetrical sprocket?
  - Delete sprocket adapter?

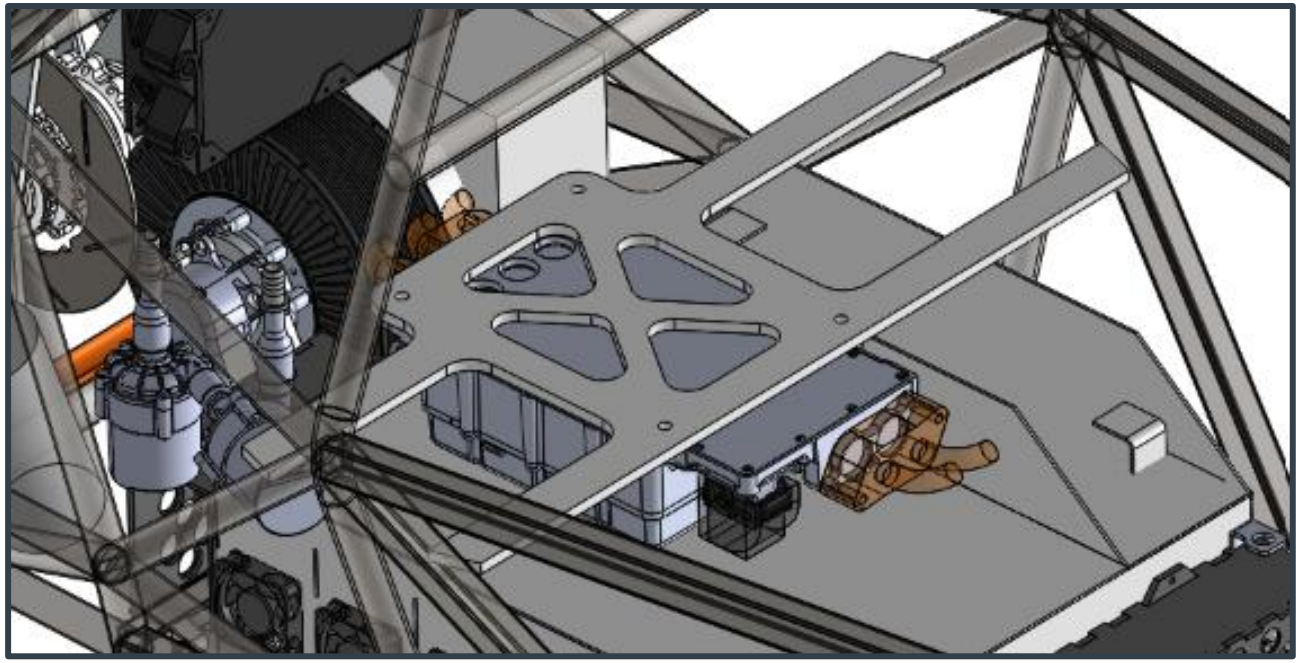


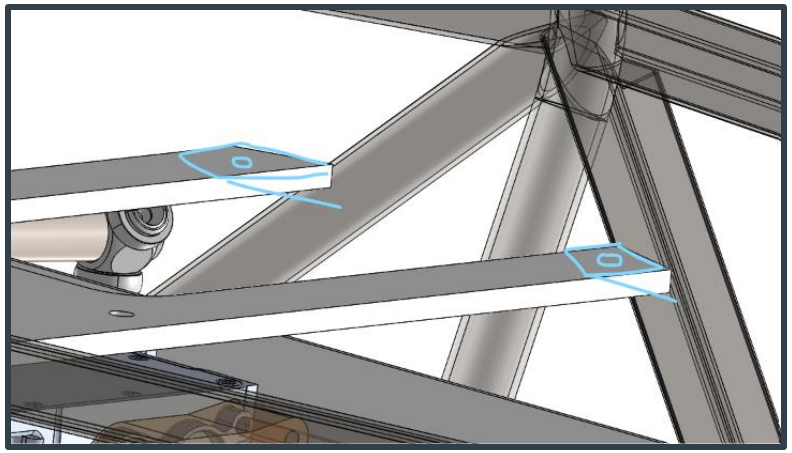
Pocketing Experimentation

0.12992126 bolt radius, 8 bolts
(from last year, plan to look at alternative options)

Inverter mounting plan

- **CARBON FIBER**
- Upside down flip improves CG and mounting
- As far back as possible for rearward CG target goal





3 Installing the Inverter

All CM inverters have 4 mounting locations, one at each corner. Mounting orientation is not critical. The inverter should be mounted in a location that is not exposed to direct spray from water. Each mounting hole is sized to handle up to a M10 socket head cap screw. For dimensional drawings refer to the product page on the Cascadia Motion website.



The University of Texas at Austin
Cockrell School of Engineering